Estimation and Learning in Aerospace 2025/2026 exam project

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Objective

- Identification of the dynamics of a small-scale UAV requires the ability to measure outputs such as angular rate and acceleration components in the body frame.

- In this project we focus on a single axis of the drone, with the following objectives:
  - Identify a grey-box model for the longitudinal dynamics of the drone using the baseline measurement set up.
  - Carry out a sensitivity analysis to perturbations in measurement noise properties.
  - Select the best model and carry out a Monte-Carlo study to assess its uncertainty.

Objectives

Project tasks

Task 1

- System identification of longitudinal dynamics from simulated data, using the frequency-domain grey-box method

Task 2

- Sensitivity analysis to perturbations in measurement noise properties

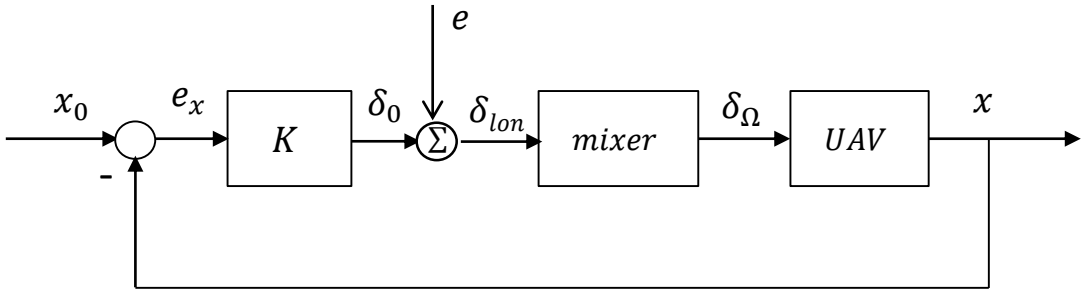
Task 3

- Analysis of the best model in a Monte Carlo study

A small quadrotor

Weight: 270g
Dimensions: 20 x 20 x 4 cm
Diameter (motor-to-motor distance): 16cm
Hovering time: 7’30’’

Identification experiments and I/O estimation data



Decoupled dynamics

- *Longitudinal dynamics*
  - Input: δ_{lon}/ Outputs: $q,\ a_x$

Linearised model for longitudinal dynamics

State equation

$$\begin{bmatrix} \dot{u} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_q & -g \\ M_u & M_q & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} X_\delta \\ M_\delta \\ 0 \end{bmatrix} \delta_{lon}$$

u Longitudinal (body) velocity [m/s]
q Pitch rate [rad/s]
θ Pitch angle [rad]
a_x Longitudinal (body) acceleration [m/s^2]
δ_{lon} Pitch moment [normalised to -1, +1 range]

Output equation

$$\begin{bmatrix} q \\ a_x \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ X_u & X_q & 0 \end{bmatrix} \begin{bmatrix} u \\ q \\ \phi \end{bmatrix} + \begin{bmatrix} 0 \\ X_\delta \end{bmatrix} \delta_{lon}$$

Model parameters: $\Theta = \begin{bmatrix} X_u \\ X_q \\ M_u \\ M_q \\ X_\delta \\ M_\delta \end{bmatrix}$

Linearised model for longitudinal dynamics

Model class: $M(\Theta)$

True system: $M(\Theta^*)$

$$\begin{bmatrix} \dot{u} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_q & -g \\ M_u & M_q & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} X_\delta \\ M_\delta \\ 0 \end{bmatrix} \delta_{lon}$$

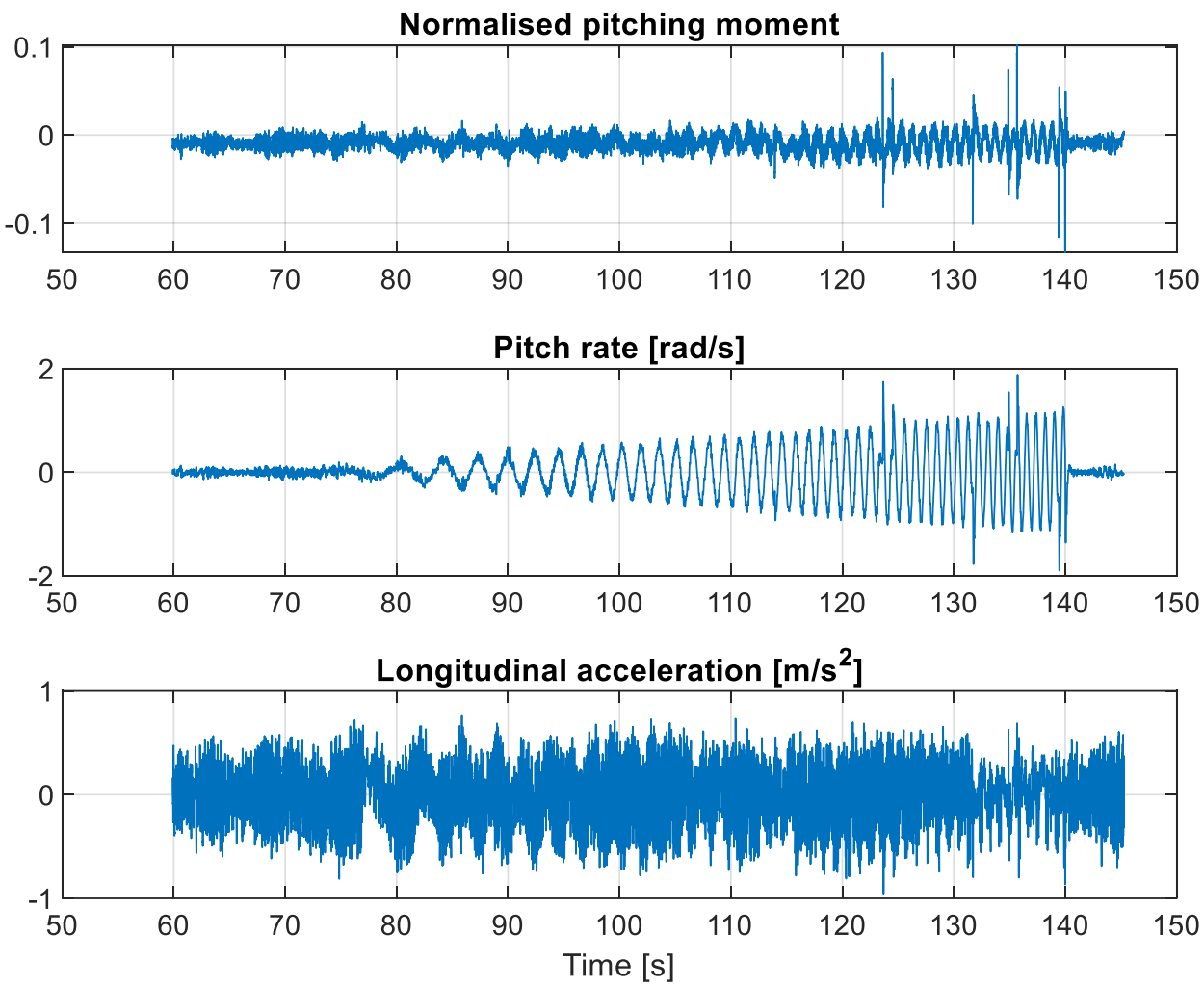
$$\begin{bmatrix} q \\ a_x \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ X_u & X_q & 0 \end{bmatrix} \begin{bmatrix} u \\ q \\ \phi \end{bmatrix} + \begin{bmatrix} 0 \\ X_\delta \end{bmatrix} \delta_{lon}$$

$$\Theta = \begin{bmatrix} X_u \\ X_q \\ M_u \\ M_q \\ X_\delta \\ M_\delta \end{bmatrix}$$

Longitudinal dynamics: excitation signal

- Experiments are carried out under position and attitude feedback, in order to guarantee safe operation.

- Baseline excitation input is a long duration sweep: excite low-frequency translational dynamics



Simulator

The simulator is representative of the local linear closed-loop dynamics of the drone and combines the identified model with a complete model of the full attitude and position control system.

It includes the following files:

- main_quad_ANTX.m (which calls parameters_controller.m): set up the parameters and run the simulation
- ExcitationM.mat contains the vectors of time and normalised pitching moment corresponding to the baseline excitation
- Simulator_Single_Axis.slx the Simulink model corresponding to the closed-loop longitudinal dynamics

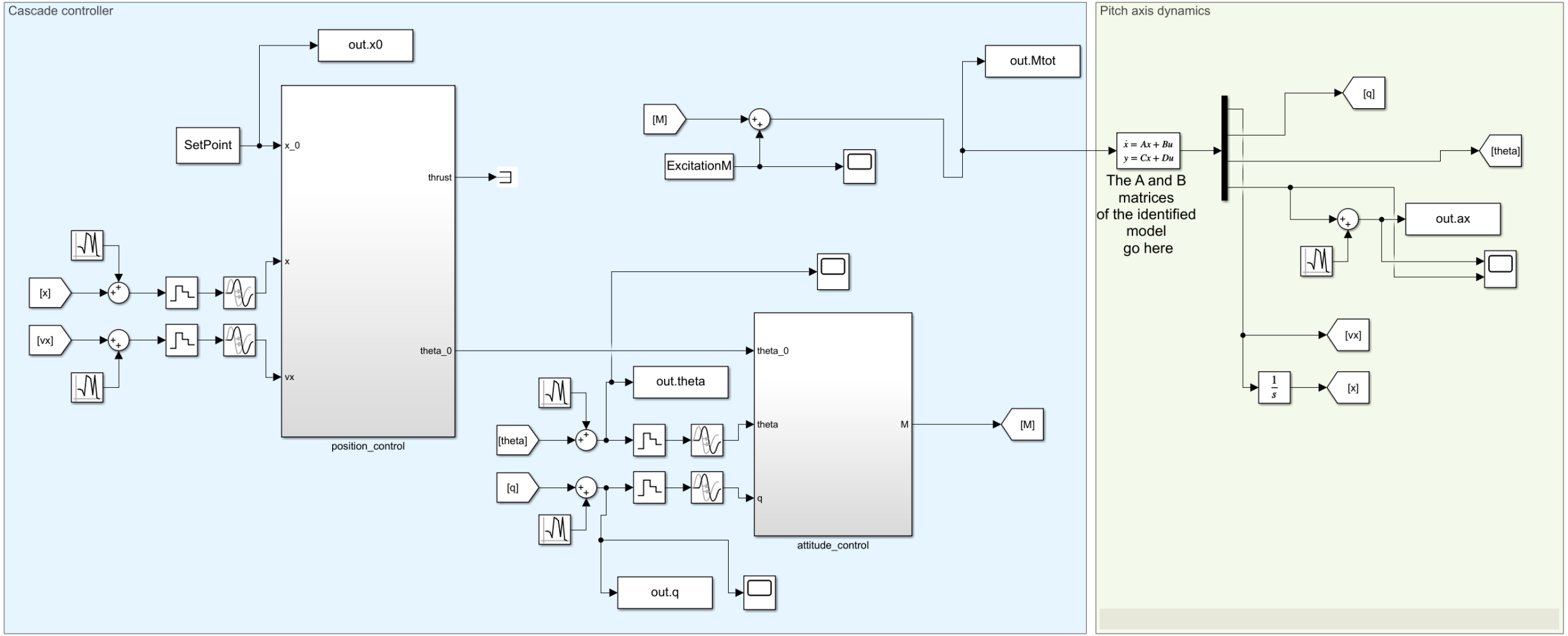
Simulator

The key parameters to model noise properties of the sensors are

- noise.ang_rate_stand_dev
- noise.acc_stand_dev

both defined in main_quad_ANTX.m.

Simulator


Cascade controller
out.x0
SetPoint
x_0
thrust
x
vx
theta_0
position_control
[x]
[vx]
[M]
ExcitationM
out.Mtot
out.theta
[theta]
[q]
theta_0
theta
q
M
attitude_control
[M]
out.q
Pitch axis dynamics
$\dot{x} = Ax + Bu$
$y = Cx + Du$
The A and B matrices of the identified model go here
[q]
[theta]
out.ax
[vx]
$\frac{1}{s}$
[x]

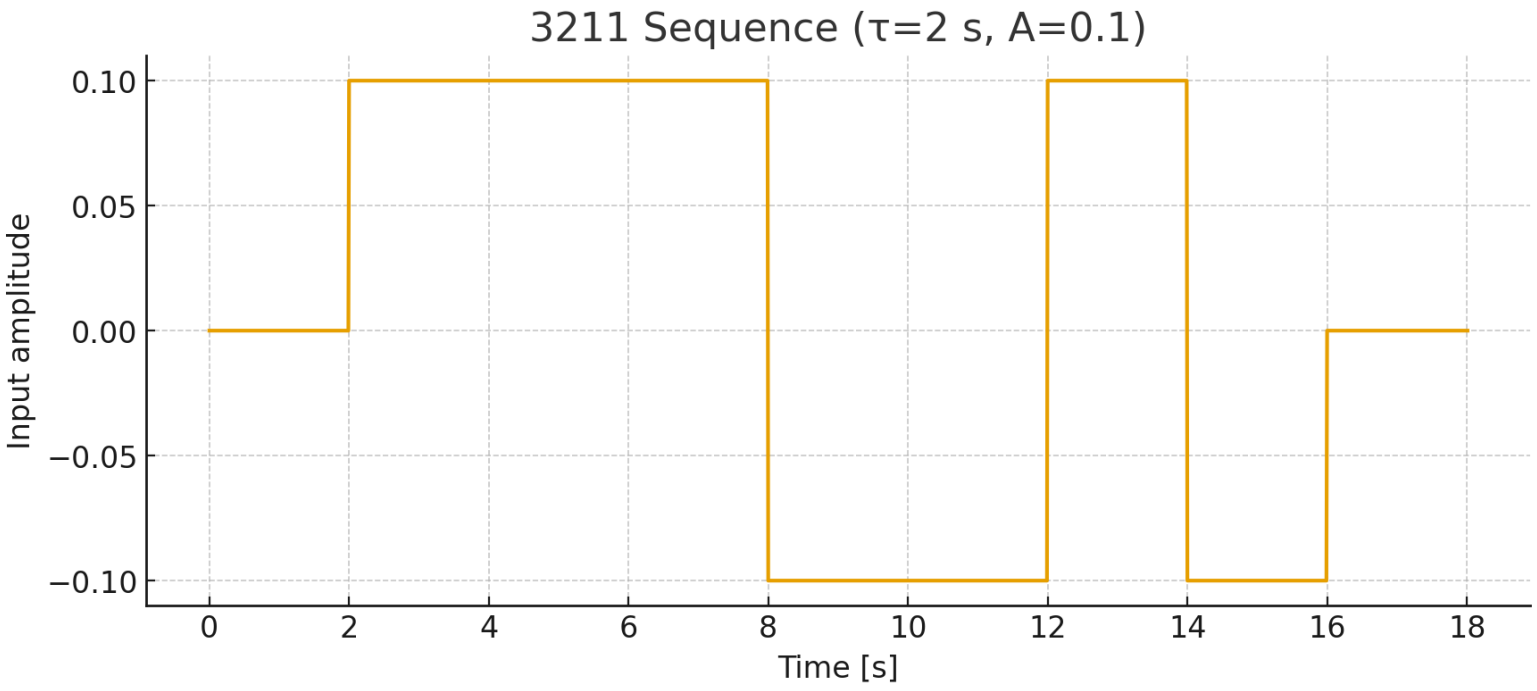
Task 1: grey-box model identification

Use the simulator and the provided baseline excitation to:

- T1.1
  - Identify a grey-box model for the longitudinal dynamics of the multirotor (input: total pitching moment; outputs: pitch rate and longitudinal acceleration) using the response to the sweep provided in the ExcitationM.mat file
- T1.2
  - Assess the uncertainty of the identified model in terms of asymptotic variance of the parameter estimates

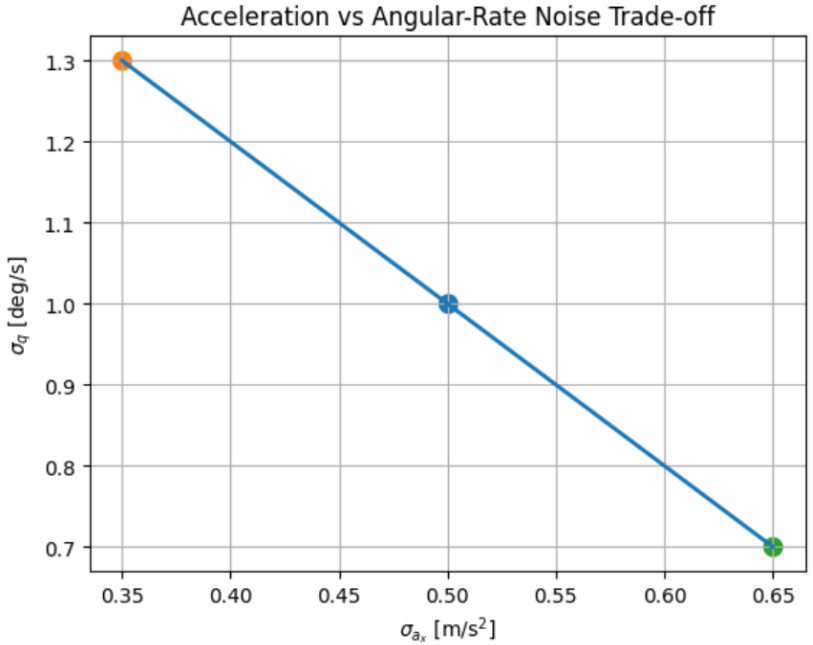
Task 1: grey-box model identification

- T1.3
  - Implement the following 3211 sequence and use it to validate the identified model



Task 2: Sensitivity analysis to perturbations in measurement noise properties

- T2.1
  - The standard deviations of the sensors are modified as in the plot by a different tuning of of the sensors mount:



  - Repeat tasks T1.1-T1.3 for each of the two perturbed conditions.

Task 2: Sensitivity analysis to perturbations in measurement noise properties

- T2.2
  - Define a criterion to select the measurement setup leading to the best model and motivate the choice
- T2.3
  - Assess the uncertainty of the best identified model in terms of asymptotic variance of the parameter estimates

Task 3: analysis and Monte Carlo study

- T3.1
  - Execute T1.1-T1.3 to identify *N* models for the longitudinal dynamics with the best setup for the sensors, using the ExcitationM.mat file for excitation input and changing the realisations of the noise processes
  - Justify the choice of *N* appropriately

- T3.2
  - Carry out a statistical analysis of the set of identified models in terms of
    - Hystograms of estimated parameters
    - Hystograms of poles and zeros
    - Dispersion of the corresponding frequency response function